

Closed loop control design for the sense mode of micromachined vibratory gyroscopes

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This paper presents a novel design method of force rebalance control for the sense mode of micromachined vibratory gyroscopes. Specific theoretical deductions are performed to identify a precise linear model of the open loop system of the sense mode, which is crucial for the PI controller design. The frequency responses obtained by experimental tests agree well with those calculated with the theoretical model, indicating the accuracy of the theoretical analyses. Experimental results demonstrate that the bandwidth of the closed loop is extended to 94.8 Hz from 2.3 Hz in the open loop and the quadrature signal is suppressed by about 64 dBV in the closed loop system. The overshoot and stable time in the step response of the closed loop system are measured to be about 15% and 35 ms, respectively. The mode-splitting gyroscope with the closed loop controlled sense mode achieves a scale factor of 41.0 mV/deg/s with nonlinearity of 0.09% and asymmetry of 1%, and a bias instability of 4.0 °/h with angle random walk of 0.171 deg/h^{1/2}.

MEMS vibratory gyroscope, sense mode, closed loop control, linear model

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1 Introduction

Micromachined vibratory gyroscope is a must for more and more intelligent fields, and the vast application market makes it significant to further improve the gyroscope's performances [1, 2]. In order to obtain stable amplitude of the drive velocity, a series of mature closed loop control methods have been adopted for the drive mode [3–5]. As for the sense mode, open loop control is a prevalent method of many works for its simplicity [3–8]. However, there are some limitations resulted from this method. For instance, it is hard to control the bandwidth, nullify the quadrature error and extend the measurement range [7, 8]. Furthermore, it is difficult to suppress the drift of those parameters induced by

the variations of ambient temperature and pressure [9]. Therefore, closed loop control for the sense mode should be adopted for high precision application demands.

Nowadays, sigma-delta modulator loops are widely used to realize the force feedback control [10–12]. It can directly output digital stream signals that are prevailed for integration and post-digital processing. Nevertheless, the robustness and stability are still problems since the signal-noise-ratio (SNR) and stability may decrease due to the nonlinearity of the quantizer as the input amplitude increases [13, 14]. Moreover, the sigma-delta loop will become more complicated when a mechanical element is present in the closed loop.

Another method for closed loop control of the sense mode is based on demodulation and modulation [15, 16]. In order to utilize the classical linearity control theory for the controller design, a precise linear model for the controlled

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object $G(s)$ is vital to implement the force rebalance control. Whereas, in those approaches, the analytical model that can be applied to both matched-mode and split-mode gyroscopes are not sufficiently studied. In ref. [16], the transfer function of the open loop system $G(s)$ had to be identified with Levy method based on the experimental results, which is time-consuming and may lead to inaccuracy of the model. In addition, the difference of the two resonant frequencies and the quality factor of the sense mode may vary with ambient temperature, but the center frequency and the quality factor of the notch filter are fixed, which will affect the stability of the closed loop seriously.

In this work, the linear model of $G(s)$ is analytically derived and no notch filter is used in the control loop, meanwhile, the controller is also simplified. The accuracy of the theoretical model and the novel control method has been verified experimentally.

2 Theoretical analyses

2.1 System overview

The structure design aims to make a gyroscope achieve a sub 10 deg/h bias instability, as well as a bandwidth larger than 90 Hz. Figure 1 shows the simplified schematic of a Z-axis doubly decoupled MEMS vibratory tuning fork gyroscope. There are two groups of combs in the drive mode, namely slide-film drive combs and slide-film drive-sensing combs. Simultaneously, there are two groups of combs in the sense mode, namely squeeze-film sense combs and slide-film force feedback combs. The drive combs and drive-sensing combs are used for closed loop control of the drive mode while the sense combs and force feedback combs are designed for the closed loop control of the sense mode.

The force feedback control system for the sense mode of

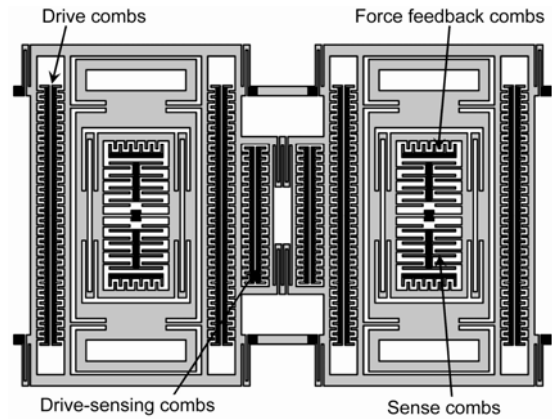


Figure 1 The simplified schematic of a Z-axis doubly decoupled MEMS tuning fork gyroscope.

a gyroscope is shown in Figure 2. There are two closed loops in the system due to the demodulation and remodulation processes, namely Coriolis force rebalance control loop and quadrature rebalance control loop. Given the system's completeness, the demodulation and remodulation signals of the two loops should be orthogonal. After the G-PI control, a remodulation process is required to construct the feedback force to balance the external forces. $y_i(t)$ and $y_q(t)$ are then used as the final outputs of the closed loop system.

Different from the schematic proposed in ref. [16], there is no notch filter in the control loops and the two loops are similar to each other, which can not only guarantee the same bandwidth of them but also simplify the design of the controllers. Taking the Coriolis force rebalance control loop for example, in order to utilize the classical linearity control theory for the G-PI controller design, the closed loop can be simplified to an equivalent linear system, as shown in Figure 3. The linear model $G(s)$ represents the response from the angular rate input to the output of the low pass filter (LPF) shown in Figure 2.

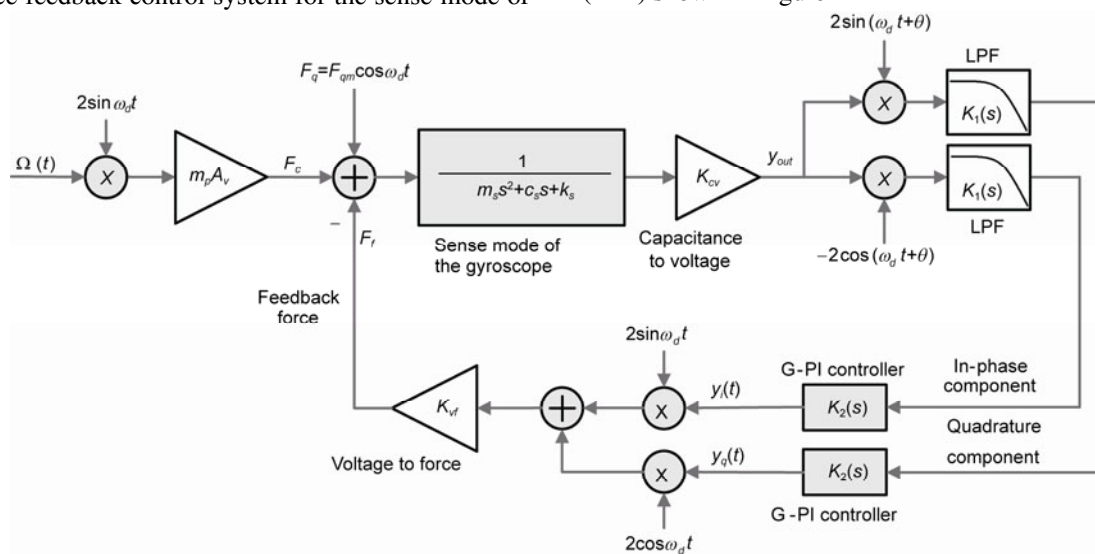


Figure 2 Block diagram of the force rebalance control system for the sense mode of a gyroscope.

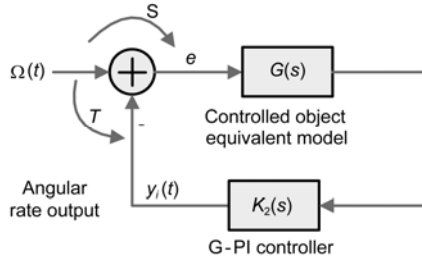


Figure 3 Equivalent linear system of the force feedback control for in-phase component of the sense mode.

2.2 Linear model identification

It is crucial to identify the controlled object $G(s)$ because the G-PI controller should be designed based on it. A precise analytical model is much preferred to provide important guidance for the experiments. Assume that the angular rate and the velocity of the drive mode are as follows

$$\Omega(t) = \Omega_R \cos(\omega_R t), \quad (1)$$

$$v(t) = A_v \sin(\omega_d t), \quad (2)$$

where ω_d and ω_R represent the resonant frequency of the drive mode and the frequency of angular rate, respectively. The dynamical equation of the sense mode of a vibratory gyroscope can be expressed as

$$m_s \ddot{y} + c_s \dot{y} + k_s y = -2m_p \Omega v, \quad (3)$$

where m_s and m_p are the sense mass and proof mass of the gyroscope, respectively. c_s , k_s and y are the damping coefficient, stiffness coefficient and the displacement of the sense mode, respectively. The Coriolis force can be written as

$$F_c = -2m_p \Omega v = -F_{cm} \cos(\omega_R t) \sin(\omega_d t), \quad (4)$$

where $F_{cm} = 2m_p A_v \Omega_R$ is the amplitude of the Coriolis force. Then, the output y_{out} can be obtained as

$$y_{out} = \frac{F_{cm} k_{cv}}{2m_s} [A_1 \sin(\omega_d + \omega_R)t + B_1 \cos(\omega_d + \omega_R)t + A_2 \sin(\omega_d - \omega_R)t + B_2 \cos(\omega_d - \omega_R)t], \quad (5)$$

where ω_R is a common variable of A_1 , A_2 , B_1 and B_2 , as shown in eqs. (6)–(9). ω_s and Q_s represent the resonant frequency and quality factor of the sense mode, respectively. K_{cv} is the gain coefficient from capacitance to voltage while K_{vf} is the conversion coefficient from voltage to force.

$$A_1 = \frac{\omega_s^2 - (\omega_d + \omega_R)^2}{[\omega_s^2 - (\omega_d + \omega_R)^2]^2 + \omega_s^2 (\omega_d + \omega_R)^2 / Q_s^2}, \quad (6)$$

$$B_1 = \frac{-\omega_s (\omega_d + \omega_R) / Q_s}{[\omega_s^2 - (\omega_d + \omega_R)^2]^2 + \omega_s^2 (\omega_d + \omega_R)^2 / Q_s^2}, \quad (7)$$

$$A_2 = \frac{\omega_s^2 - (\omega_d - \omega_R)^2}{[\omega_s^2 - (\omega_d - \omega_R)^2]^2 + \omega_s^2 (\omega_d - \omega_R)^2 / Q_s^2}, \quad (8)$$

$$B_2 = \frac{-\omega_s (\omega_d - \omega_R) / Q_s}{[\omega_s^2 - (\omega_d - \omega_R)^2]^2 + \omega_s^2 (\omega_d - \omega_R)^2 / Q_s^2}. \quad (9)$$

The displacement output y_{out} is demodulated by a signal of $-2\cos(\omega_d t + \theta)$ and then passes through an LPF to remove the high-frequency components, as shown in Figure 2. The in-phase component is thus described as

$$y_0 = -\Omega_R (A \sin \omega_R t + B \cos \omega_R t), \quad (10)$$

where

$$A = \frac{m_p A_v k_{cv}}{m_s} (A_1 \cos \theta - A_2 \cos \theta + B_1 \sin \theta - B_2 \sin \theta), \quad (11)$$

$$B = \frac{m_p A_v k_{cv}}{m_s} (B_1 \cos \theta + B_2 \cos \theta - A_1 \sin \theta - A_2 \sin \theta). \quad (12)$$

Hence, from the Laplace transforms of eqs. (1) and (10), the transfer function of the in-phase component can be derived as

$$SF_{open}(s) = -B + Aj = -B + As / \omega_R. \quad (13)$$

When taking the LPF into account, the linear model of the open loop $G(s)$ can be derived as

$$G(s) = SF_{open}(s) \times K_1(s) = (-B + Aj) \times \frac{\omega_0^2}{s^2 + 2\xi_0 \omega_0 s + \omega_0^2}, \quad (14)$$

where ω_0 and ξ_0 are the cut-off frequency and damping ratio of the second-order low pass filter $K_1(s)$, respectively and $s = j\omega_R$. Therefore, the controlled object $G(s)$ presented by eq. (14) can be used to design the G-PI controller.

The sensitivity is usually contradictory to the bandwidth, the smaller the splitting frequency, the higher the sensitivity, and the smaller the bandwidth. The quality factor of the sense mode also affects the bandwidth. In terms of open loop control of the sense mode, some gyroscope designers used to make a tradeoff and set the difference of the two resonant frequencies to be between 100 and 300 Hz to guarantee an adequate bandwidth. However, it is at the cost of sensitivity even though Q_s is very high. Fortunately, when it comes to closed loop control, the bandwidth can be tuned easily, which helps to solve the contradiction. In this paper, the splitting frequency is set only 8.1 Hz to enhance the sensitivity, meanwhile, the bandwidth can be adjusted accordingly by the closed loop control. The main parameters of the gyroscope we used are shown in Table 1.

2.3 Generalized PI controller design

Different from the controllers presented in ref. [16], there is no notch filter applied to the control loop, and a simple

Table 1 Parameters of the vacuum packaging gyroscope

Parameters	Value
ω_d (rad/s)	$2\pi \times 8355.1$
Q_d	2470
ω_s (rad/s)	$2\pi \times 8347$
Q_s	1620

generalized PI controller (G-PI) is adopted in the feedback loop, as depicted in eq. (15). $(s+\omega_2)/(s+\omega_1)$ can be regarded as a PI controller, and $k/(s+\omega_3)$ is a first-order low pass filter used to eliminate the high frequency components. The simplification of the controller not only reduces the complication of realization but also enhances the reliability of the control system.

$$K_2(s) = \frac{k(s+\omega_2)}{(s+\omega_1)(s+\omega_3)}. \quad (15)$$

In order to simplify the analysis, based on Figures 2 and 3, we can assume that K_{vf} is equal to $m_p A_v$, then the transfer function of the open loop $L(s)$ can be derived as

$$L(s) = y_i(s)/\Omega(s) = G(s) \times K_2(s). \quad (16)$$

Thus, the closed loop transfer function $T(s)$ and sensitivity transfer function $S(s)$ can be defined as follows

$$T(s) = L(s) / (1 + L(s)), \quad (17)$$

$$S(s) = 1 / (1 + L(s)). \quad (18)$$

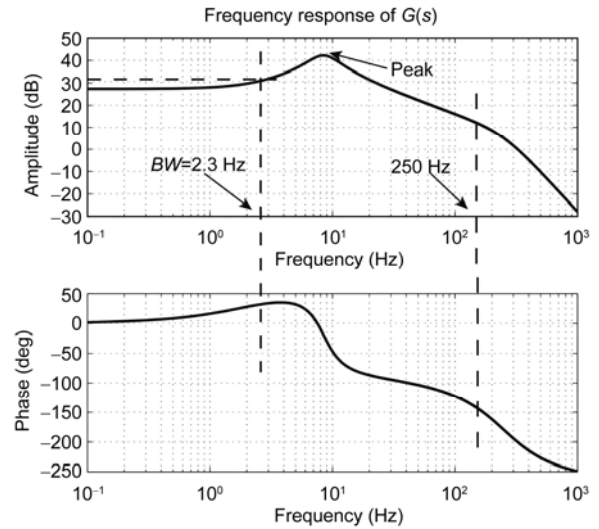
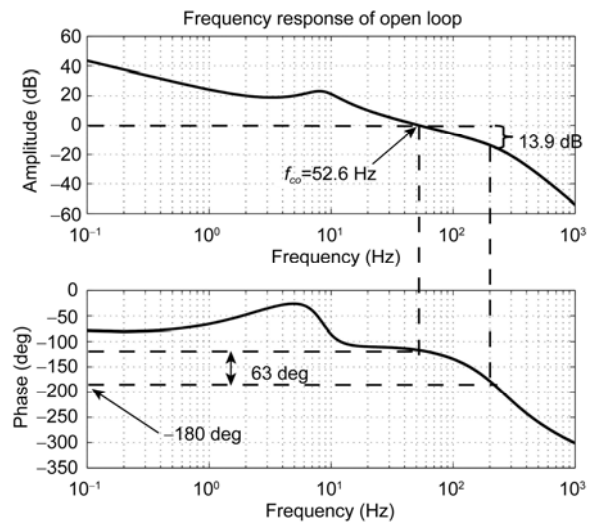
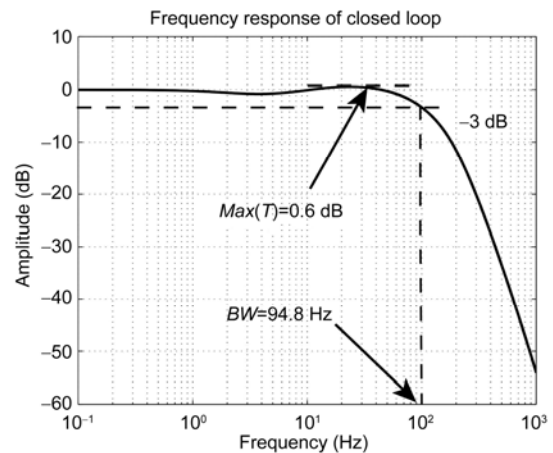
So far, the frequency-domain simulations can be implemented with the theoretical eqs. (14)–(18).

3 Simulation in the frequency domain

The frequency responses of $G(s)$, $L(s)$, $T(s)$ and $S(s)$ can be simulated with the analytical linear models above. The simulation results calculated with Matlab are illustrated in Figures 4, 5, 6 and 7, respectively.

The frequency response of $G(s)$ is shown in Figure 4. In order to eliminate the high frequency signals after multiplication demodulation and affect the phase response less, the cut-off frequency of the LPF is set to be 250 Hz. The splitting frequency is 8.1 Hz, resulting in a 2.3 Hz final bandwidth of the open loop system. In addition, there is a peak located at the difference of the two resonant frequencies, as shown in the amplitude-frequency response of $G(s)$, but the phase is not affected seriously, therefore, there is no need to apply a notch filter to compensate the phase, which will avoid those problems induced by the notch filter.

After the G-PI control, the frequency responses of $L(s)$, $T(s)$ and $S(s)$ are depicted in Figures 5, 6 and 7, respectively. The crossover frequency is set to be 52.6 Hz to obtain enough phase margin and bandwidth, as shown in Figure 5.

**Figure 4** Frequency response of $G(s)$.**Figure 5** Frequency response of the open loop $L(s)$.**Figure 6** Frequency response of the closed loop $T(s)$.

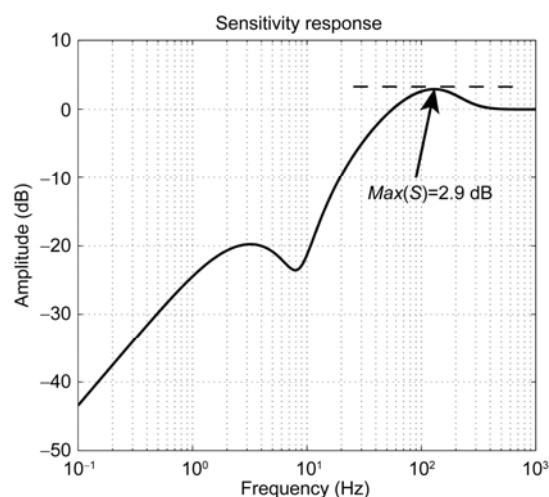


Figure 7 Frequency response of the sensitivity transfer function $S(s)$.

The phase margin PM , gain margin GM , maximum sensitivity M_s and maximum complementary sensitivity M_t are calculated to be 63° , 13.9, 2.9 and 0.6 dB, respectively, which guarantees the stability and robustness of the closed loop system. Meanwhile, the bandwidth of the closed loop system is 94.8 Hz, as shown in Figure 6. Due to the adjustment by G-PI controller, the bandwidth has been extended to 94.8 Hz from 2.3 Hz of the open loop, which can satisfy more application demands.

4 Experimental tests

After the simulation by Matlab, the relevant parameters of the G-PI controller or other modules can be confirmed, thereby, the closed loop control system can be realized in experiments according to the theoretical analysis. Figure 8 shows the test circuit for the closed loop control system. This circuit mainly comprises a closed loop for the drive mode and a closed loop for the sense mode. The vacuum sealed gyroscope is mounted on the PCB. In the experimental tests, a virtual rate-table method [17] was adopted to

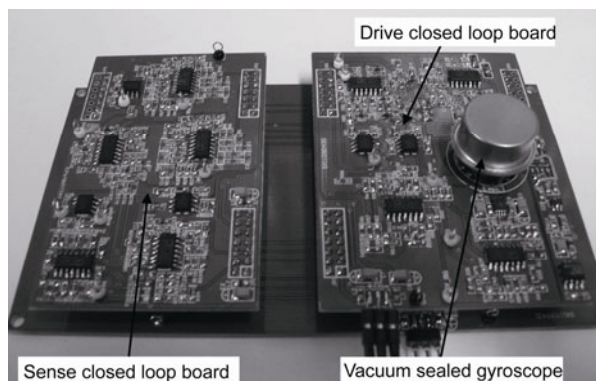


Figure 8 Test circuits for the closed loop control system.

obtain the corresponding frequency responses.

The frequency response test of the open loop system and closed loop system are illustrated in Figure 9. The dash line represents the response of $G(s)$ while the solid line stands for that of $T(s)$. For comparing convenience, an offset was added to the $G(s)$ so that the starting point was 0 dB, which did not affect the calculation of the bandwidth. From Figure 9, the bandwidths of the open loop system and closed loop system are measured to be 2.3 and 94.8 Hz, respectively, which accord with the simulation results above.

The coincidence between the experimental and simulation results verifies the accuracy of the theoretical analysis. Now that the controlled object $G(s)$ can be obtained precisely by theoretical deductions and simulations, the G-PI controller can be designed beforehand, which can vastly advance the working efficiency and supply a valuable guidance for experiments.

In order to further evaluate the performance of the closed loop system, the time-domain response test was also necessary. The step response of the closed loop system is shown in Figure 10. For comparing convenience, the angular rate

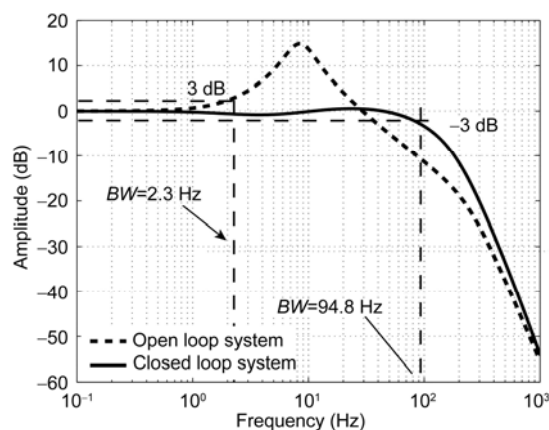


Figure 9 The frequency responses of the open loop system and closed loop system acquired by experimental tests.

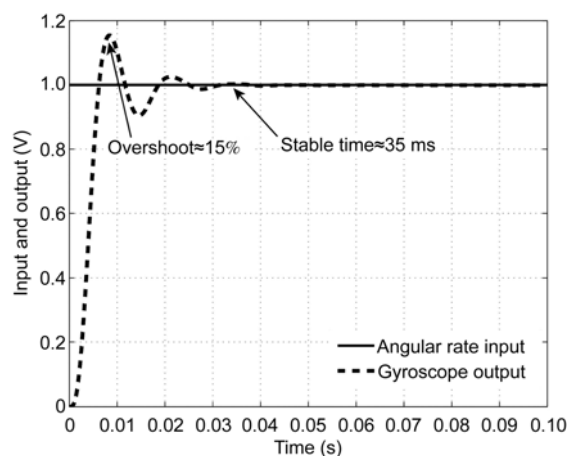


Figure 10 Step response of the closed loop system.

input signal was converted to a corresponding voltage signal with a conversion coefficient of the scale factor. The scale factor was 41.0 mV/deg/s, which induced the gyroscope output to be 1 V when the angular rate input was 24.39 deg/s. The overshoot and the stable time were about 15% and 35 ms, respectively, which indirectly testified the accuracy of the phase margin, gain margin and crossover frequency announced in the simulation results.

The amplitude of the quadrature signal was 8 V in the open loop system, while it was only about 5 mV in the closed loop system, as demonstrated in Figure 11. The suppressed factor was about 64 dB, which strongly verified the effectiveness of the quadrature rebalance control loop.

The scale factor test of the closed loop system is shown in Figure 12. The scale factor, nonlinearity and asymmetry were measured to be 41 mV/deg/s, 0.09% and 1% with the input range of $\pm 200^\circ/\text{s}$, respectively. As shown in Figure 13, the bias instability and angle random walk (ARW) of the mode-splitting gyroscope with closed loop controlled sense mode were measured to be 4.0 deg/h and 0.171 deg/h^{1/2},

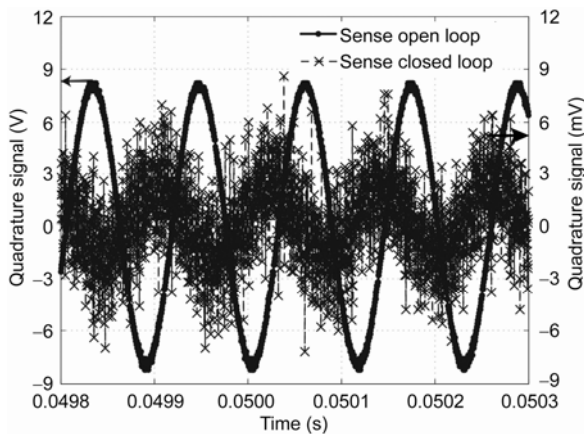


Figure 11 Quadrature signals in the open loop system and closed loop system.

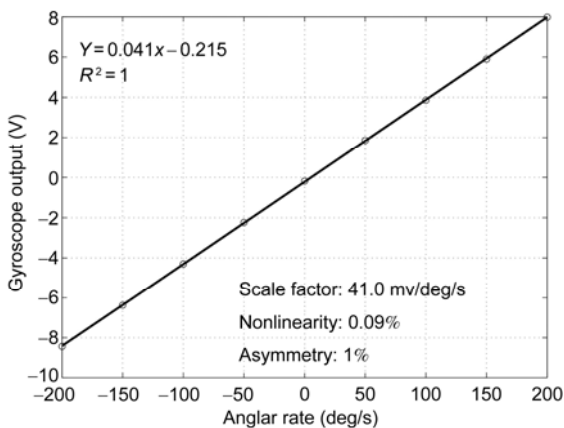


Figure 12 Scale factor test of the closed loop system.

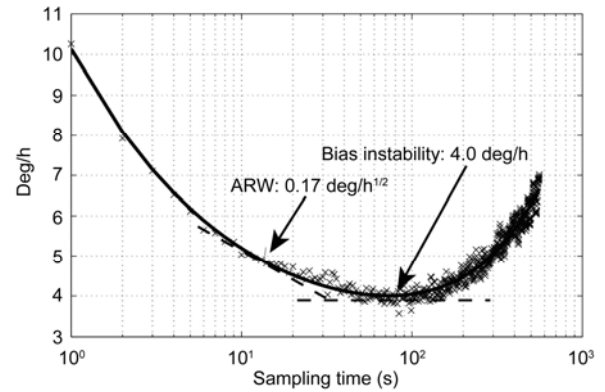


Figure 13 Allan variance graph of the mode-splitting gyroscope with closed loop controlled sense mode.

respectively, which reached the tactical level.

5 Conclusion

This paper has presented a detailed design method for the force rebalance control for the sense mode of MEMS vibratory gyroscopes. A precise linear model for the open loop of the sense mode was deduced theoretically and proved by experiments, which contributes to the design of G-PI controller. Both frequency-domain and time-domain responses verified the stability and robustness of the novel control system. Furthermore, the nonlinearity, bandwidth and the bias instability of the mode-splitting gyroscope with closed loop controlled sense mode were measured to be 0.09%, 94.8 Hz and 4.0 deg/h, respectively, which achieved the design goals set at the beginning.

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